Differentiation and Rates of Change

ANSWER KEY

Section 1: Product and Quotient Rules

1 Differentiate $y = x^2 e^{3x}$ with respect to x.

a. $xe^{3x}(3x + 2)$ ✓

b. $2xe^{3x}$

c. $3x^2 e^{3x}$

d. $x^2 e^{3x} + 2x$

Product rule: $y' = (2x)(e^{3x}) + (x^2)(3e^{3x}) = xe^{3x}(2 + 3x)$.

2 Find $\frac{dy}{dx}$ for $y = \frac{2x-1}{x+4}$.

a. $\frac{9}{(x+4)^2}$ ✓

b. $\frac{2}{(x+4)^2}$

c. $\frac{7}{(x+4)^2}$

d. $\frac{2x+9}{(x+4)^2}$

Quotient rule: $\frac{2(x+4)-(2x-1)(1)}{(x+4)^2} = \frac{2x+8-2x+1}{(x+4)^2} = \frac{9}{(x+4)^2}$.

3 Find the derivative of $f(x) = x \ln x$ evaluated at $x = e$.

a. 2 ✓

b. 1

c. e

d. 0

$f'(x) = 1 \cdot \ln x + x \cdot \frac{1}{x} = \ln x + 1$. At $x = e$: $f'(e) = \ln e + 1 = 1 + 1 = 2$.

Section 2: Chain Rule and Composite Functions

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4 Differentiate $y = (3x^2 - 5)^4$ with respect to x.

a. $24x(3x^2 - 5)^3$ ✓

b. $12x(3x^2 - 5)^3$

c. $4(3x^2 - 5)^3$

d. $18x^2(3x^2 - 5)^3$

Chain rule: $\frac{dy}{dx} = 4(3x^2 - 5)^3 \cdot (6x) = 24x(3x^2 - 5)^3$.

5 Find the derivative of $y = \sin^3(2x)$.

a. $6\sin^2(2x)\cos(2x)$ ✓

b. $3\sin^2(2x)\cos(2x)$

c. $6\sin(2x)\cos(2x)$

d. $-6\sin^2(2x)\cos(2x)$

Chain rule: $3\sin^2(2x) \cdot \cos(2x) \cdot 2 = 6\sin^2(2x)\cos(2x)$.

6 Find the gradient of $y = \sqrt{4x + 9}$ at the point where $x = 4$.

a. $\frac{2}{5}$ ✓

b. $\frac{4}{5}$

c. $\frac{1}{5}$

d. 5

$\frac{dy}{dx} = \frac{1}{2}(4x + 9)^{-1/2}(4) = \frac{2}{\sqrt{4x + 9}}$. At $x = 4$: $\frac{2}{\sqrt{25}} = \frac{2}{5}$.

Section 3: Tangents, Normals and Stationary Points

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7 Find the equation of the tangent to the curve $y = x^2 - 4x + 3$ at the point where $x = 3$.

a. $y = 2x - 6$ ✓

b. $y = 2x + 6$

c. $y = -2x$

d. $y = x - 3$

At $x = 3$: $y = 9 - 12 + 3 = 0$. $\frac{dy}{dx} = 2x - 4$. At $x = 3$, $m = 2$. Line: $y - 0 = 2(x - 3) \Rightarrow y = 2x - 6$.

8 Find the coordinates of the stationary point of $y = x^3 - 3x + 2$ in the first quadrant.

a. $(1, 0)$ ✓

b. $(1, 2)$

c. $(2, 4)$

d. $(0, 2)$

$\frac{dy}{dx} = 3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = 1$ (quadrant 1). $y = 1 - 3 + 2 = 0$. Point is $(1, 0)$.

9 Determine the nature of the stationary point of $f(x) = 2x^3 - 9x^2 + 12x$ at $x = 2$.

a. **Local Minimum** ✓

b. Local Maximum

c. Point of Inflexion

d. Vertical Asymptote

$f''(x) = 12x - 18$. At $x = 2$: $f''(2) = 24 - 18 = 6 > 0$, confirming a local minimum.

Section 4: Connected Rates of Change and Applied Optimization

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10 The radius of a circular oil spill is increasing at 0.5 m/s. At what rate is the area increasing when the radius is 10 m?

a. 10π m^2/s ✓

b. 20π m^2/s

c. 5π m^2/s

d. 100π m^2/s

$A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r\frac{dr}{dt} = 2\pi(10)(0.5) = 10\pi$ m^2/s.

11 A solid cylindrical metal rod of radius r and length h has fixed volume $V = 54\pi$. Express the surface area S in terms of r and find r minimizing S.

a. $r = 3$ ✓

b. $r = 2$

c. $r = 6$

d. $r = 4$

$V = \pi r^2 h = 54\pi \Rightarrow h = 54/r^2$. $S = 2\pi r^2 + 2\pi r(54/r^2) = 2\pi r^2 + 108\pi/r$.
$\frac{dS}{dr} = 4\pi r - 108\pi/r^2 = 0 \Rightarrow 4r^3 = 108 \Rightarrow r^3 = 27 \Rightarrow r = 3$.

12 If $y = 3x^2 - 4x$ and x increases at a rate of 2 units/s, find the rate of increase of y when $x = 5$.

a. 52 units/s ✓

b. 26 units/s

c. 30 units/s

d. 60 units/s

$\frac{dy}{dt} = \frac{dy}{dx}\frac{dx}{dt} = (6x - 4)(2)$. At $x = 5$: $(26)(2) = 52$.

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